INT3405E - Machine Learning

Lecture 1: Introduction to ML

Hanoi, 09/2026

Outline

- Motivating Examples
- Introduction to Machine Learning
- Statistics vs Machine Learning
- Machine Learning vs Deep Learning
- Applications of Machine Learning
- Types of Machine Learning
  - Supervised Learning
  - Unsupervised Learning
  - Reinforcement Learning

Amazon robot



See more: https://www.youtube.com/watch?v=HSA5Bq-1fU4

Harvesting Robot



See more: https://www.youtube.com/watch?v=5chk9Sory88

Automatic Driving - Tesla

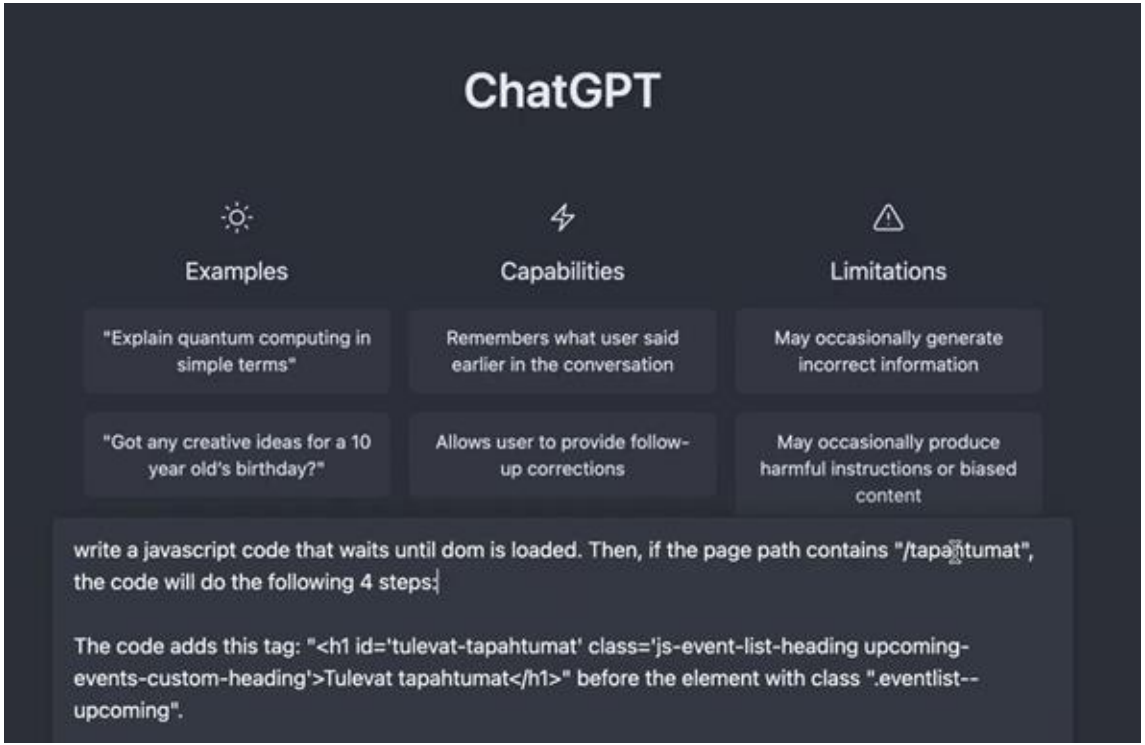


See more: https://www.youtube.com/watch?v=tlThdr3O5Qo

Youtube


5:56 / 9:55
Song:Tere Mere Milan Ki Yeh Raina Film: Abhimaan (1973) With Sinhala S...
madhukaw · 107 videos
Subscribe
1,738
35,944
41
4
Like
About
Share
Add to
Uploaded on Apr 20, 2010
No description available.
Buy "Tere Mere Milan Ki Yeh Raina" on
Ad muted. Undo
We'll do our best to show you more relevant ads in the future.
Help us show you better ads by updating your ads preferences
Google
3 Sick Bodybuilding Tips
by sixpackshortcuts
1,602,424 views
3:32
Ad
Ab To Hai Tum Sa Her Khushi Apni Iata
by malikjr1
45,661
FEATURED
4:00
Pink Interview on her husband and daughter
by MsSupergalaxy
Recommended for you
12:44
Abhimaan (1973): Loote Koi Man Ka Nagar
by tonguesonfire

ChatGPT


ChatGPT
Examples
Capabilities
Limitations
"Explain quantum computing in simple terms"
Remembers what user said earlier in the conversation
May occasionally generate incorrect information
"Got any creative ideas for a 10 year old's birthday?"
Allows user to provide follow-up corrections
May occasionally produce harmful instructions or biased content
write a javascript code that waits until dom is loaded. Then, if the page path contains "/tapahtumat", the code will do the following 4 steps:
The code adds this tag: "<h1 id='tulevat-tapahtumat' class='js-event-list-heading upcoming-events-custom-heading'>Tulevat tapahtumat</h1>" before the element with class ".eventlist--upcoming".

Beyond ChatGPT

Q1. Do Machine really Learn?

Beyond ChatGPT

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Q2. Which do prevent the real world applications of robot?

Beyond ChatGPT

Q1. Do Machine really Learn?

Q2. Which do prevent the real world applications of robot?

Q3. Can Machine solve the Millenium Prize Problems, i.e P vs NP?

What is Machine Learning?

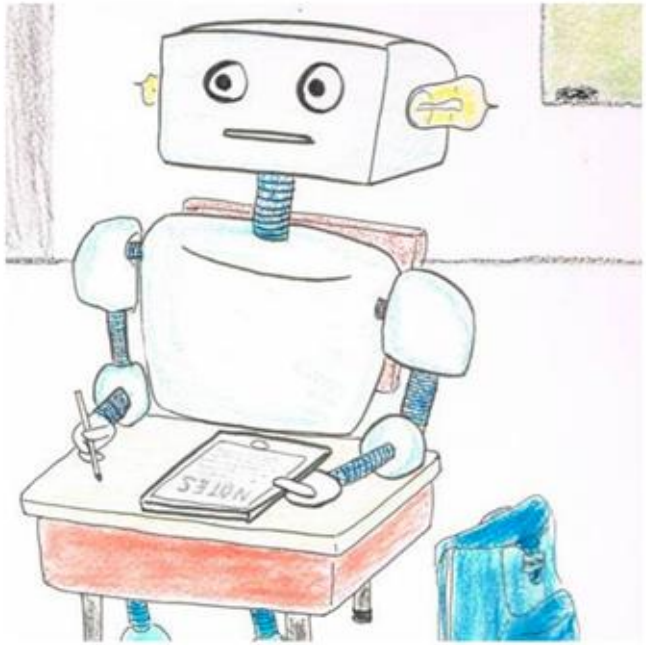


Image: www.gureckislab.org

Formal Definition of Machine Learning (ML)

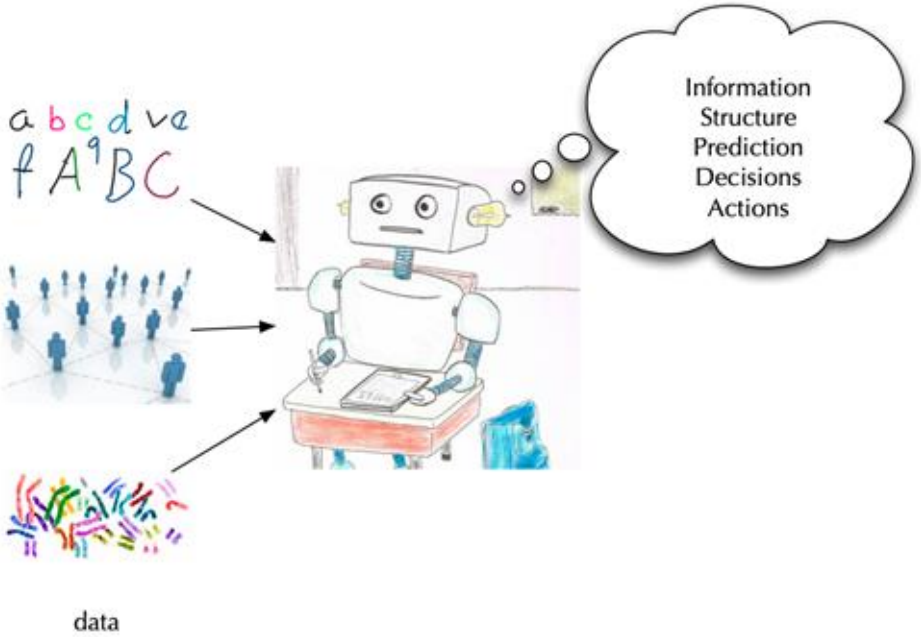


- **Arthur Samuel, 1959**: Field of study that gives computers the ability to **learn** without being explicitly programmed.
- Tom Mitchell, 1997: Any computer program that **improves its performance** at some task **through experience**.
- Kevin Murphy, 2012: To develop methods that can **automatically** detect **patterns in data**, and then to use the uncovered patterns to **predict** future data or other outcomes of interest.

Formal Definition of Machine Learning (ML)

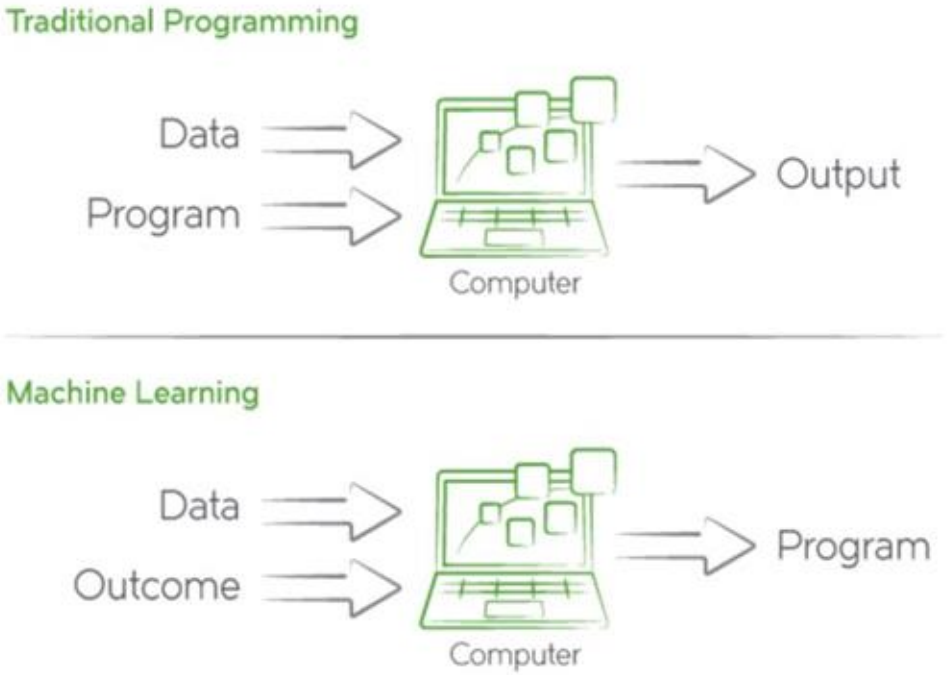
Machine learning is an application of artificial intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed.

Machine learning focuses on the development of computer programs that can access data and use it learn for themselves.



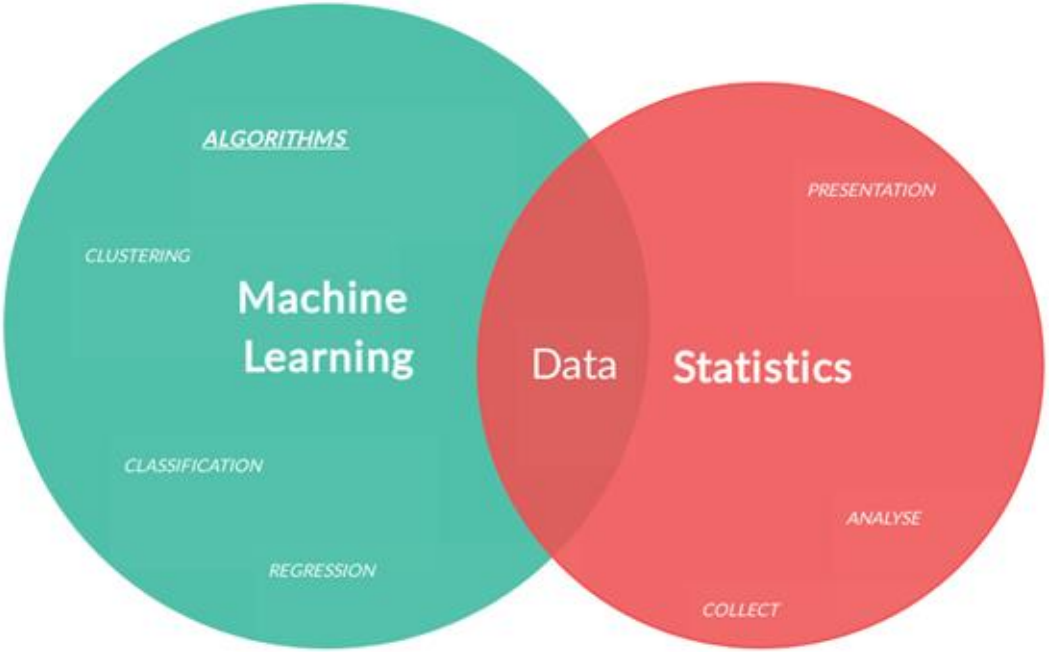
https://www.expertsystem.com/machine-learning-definition/

Traditional Programming vs Machine Learning?



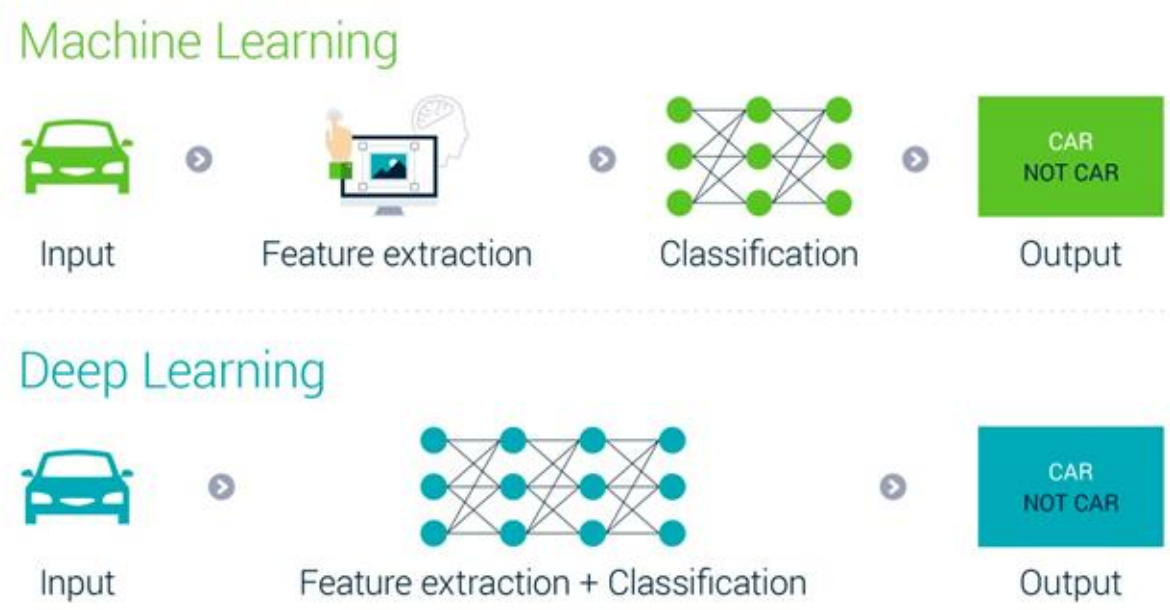
Source: https://images.techhive.com/images/article/2017/05/traditional-programming-vs-machine-learning-100723299-large.jpg

Statistics vs Machine Learning?



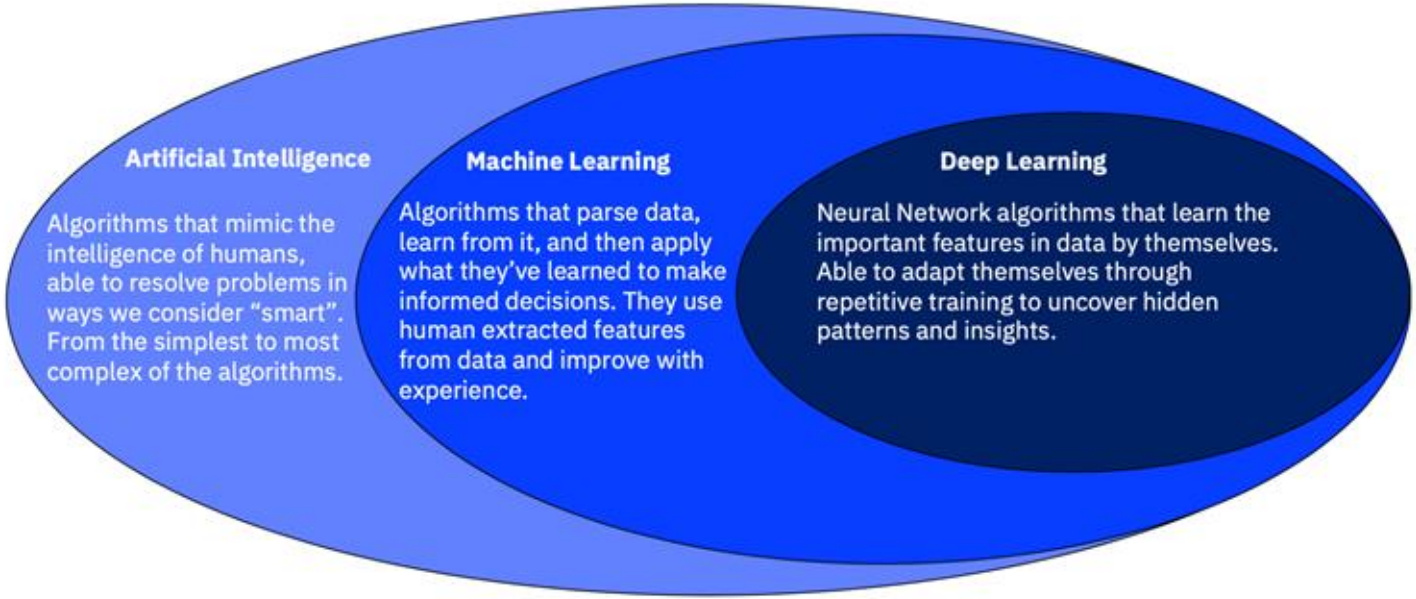
Source: https://pythonprogramminglanguage.com/what-is-the-difference-between-statistics-and-machine-learning/

Machine Learning vs. Deep Learning?



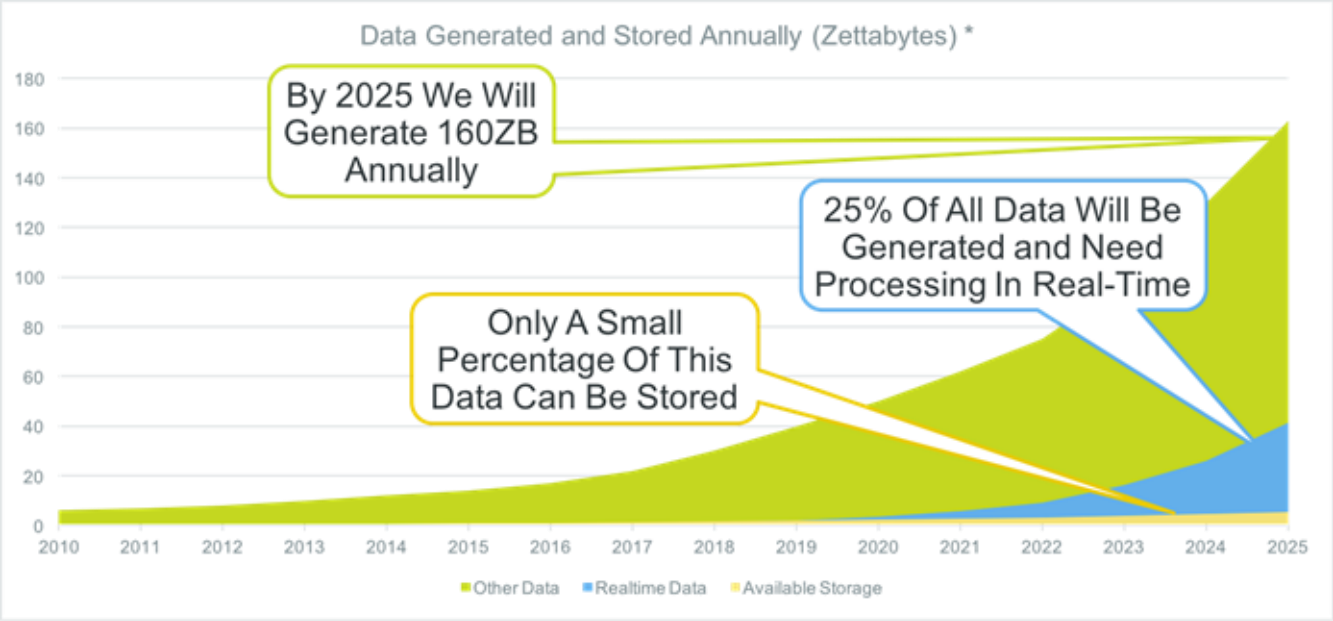
Source: https://www.linkedin.com/pulse/lets-understand-difference-between-machine-learning-vs-gauri-bapat

Machine Learning vs. Deep Learning vs AI?



Source: https://www.ibm.com/blogs/systems/ai-machine-learning-and-deep-learning-whats-the-difference/

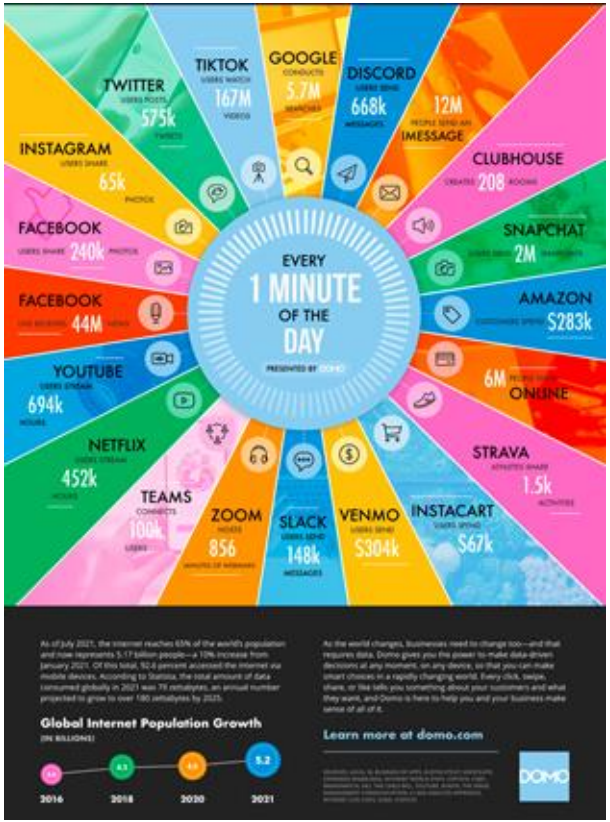
Why Machine Learning?



* Data Age 2025: The Evolution of Data to Life-Critical. An IDC White Paper, Sponsored by Seagate

https://www.striim.com/blog/real-time-data-striim-helps-prepare-for-growth/

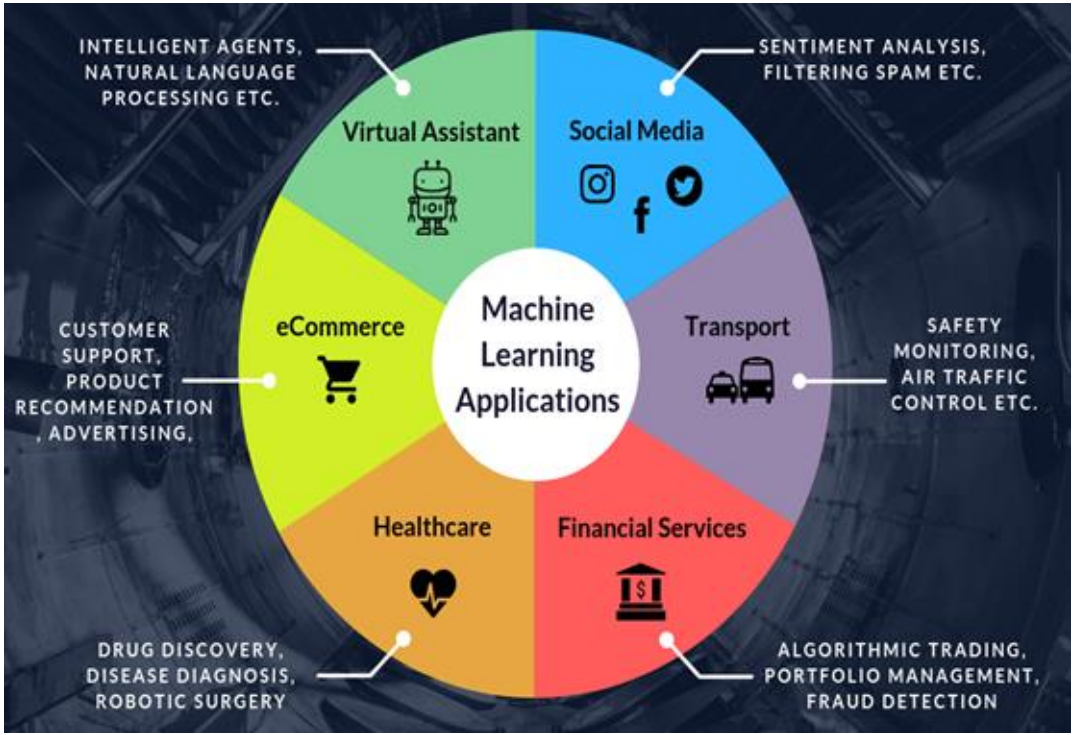
How Much Data Every Minute?



Data never sleeps

Source: https://dailyinfographic.com/how-much-data-is-generated-every-minute

Applications of Machine Learning



Source: https://www.quantinsti.com/blog/machine-learning-basics

Applications of Machine Learning: Examples



spam filtering



recommendation systems



fraud detection

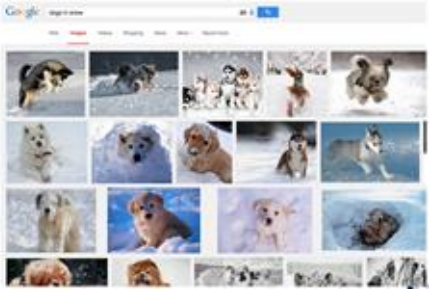


Image retrieval



self-driving cars

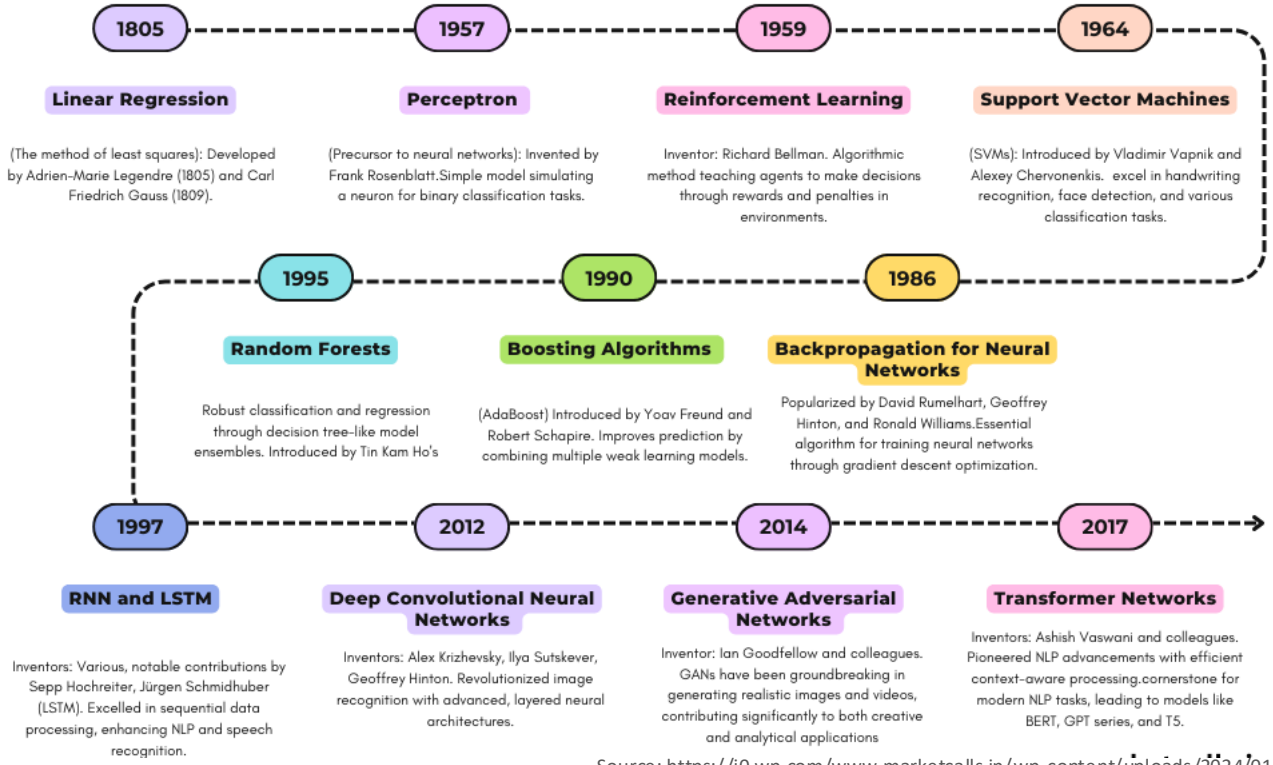


image recognition



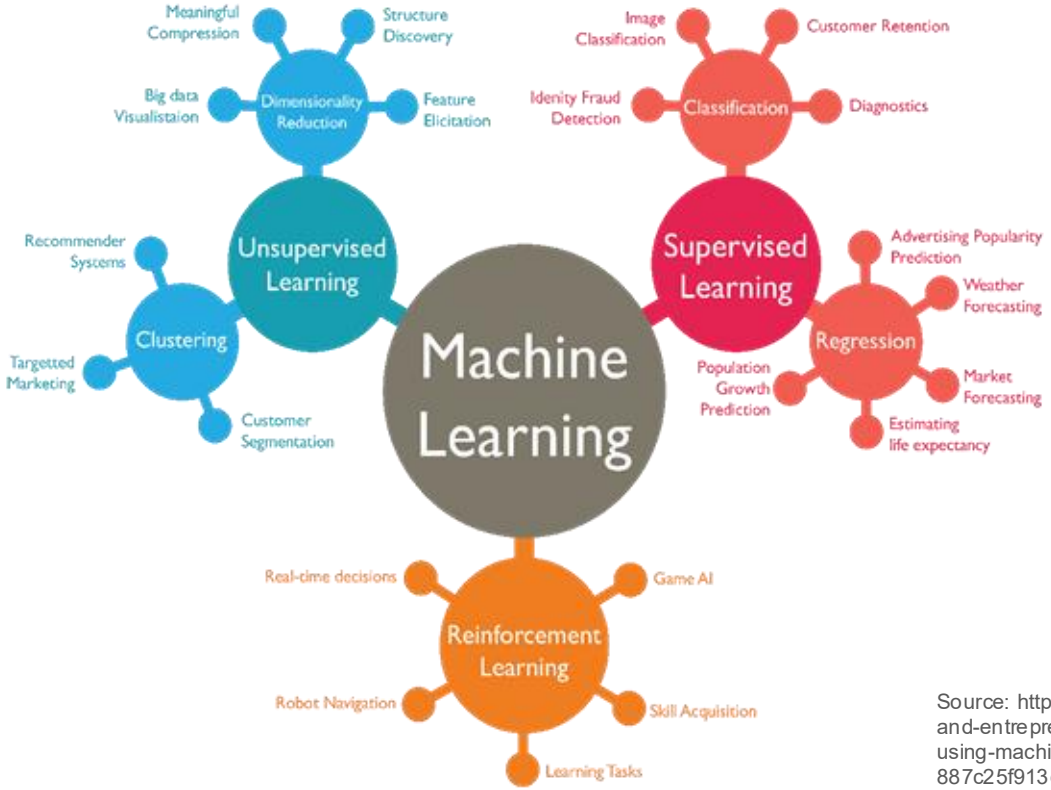
stock market analysis

A Brief History



Source: https://i0.wp.com/www.marketcalls.in/wp-content/uploads/2024/01/Evolution-of-Machine-Learning-Infographic-since-1800.png?ssl=1

Types of Machine Learning



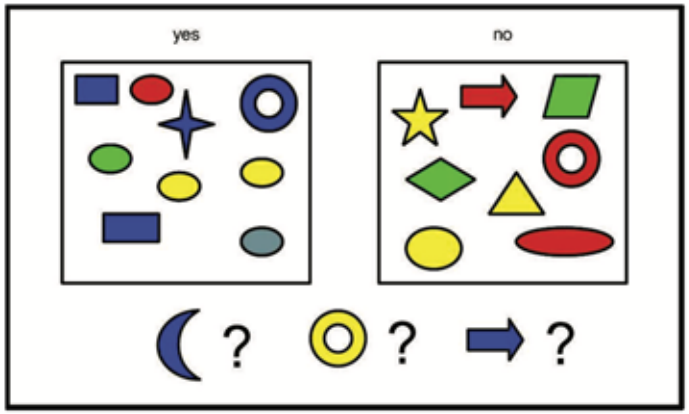
Source: https://medium.com/marketing-and-entrepreneurship/10-companies-using-machine-learning-in-cool-ways-887c25f913c3

Supervised Learning

- Given a labeled set of input-output pairs $D = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{N}$
  - D is the training set
  - N is the number of labeled examples
  - $\boldsymbol{x}$ is a d-dimensional vector of features/attributes/covariates
  - y is a response variable
- Learn a function $y = f(\boldsymbol{x})$
  - Classification when y is categorical, $y \in \{1, \ldots, C\}$
  - Regression when y is real-valued, $y \in \mathbb{R}$
- Probabilistic modeling $p(y|\boldsymbol{x}, D, \theta)$
  - θ denotes the set of model parameters

Classification

- When the response variable is categorical $y \in \{1, \ldots, C\}$
  - When $C = 2$, binary classification
  - When $C > 2$, multiclass classification
  - When $\boldsymbol{x}_i$ may have multiple labels, multi-label classification



(a)

D features (attributes)

N cases

Color	Shape	Size (cm)
Blue	Square	10
Red	Ellipse	2.4
Red	Ellipse	20.7

Label
1
1
0

(b)

Classification: Classifying Flowers



Three types of iris flowers: setosa, versicolor and virginica.

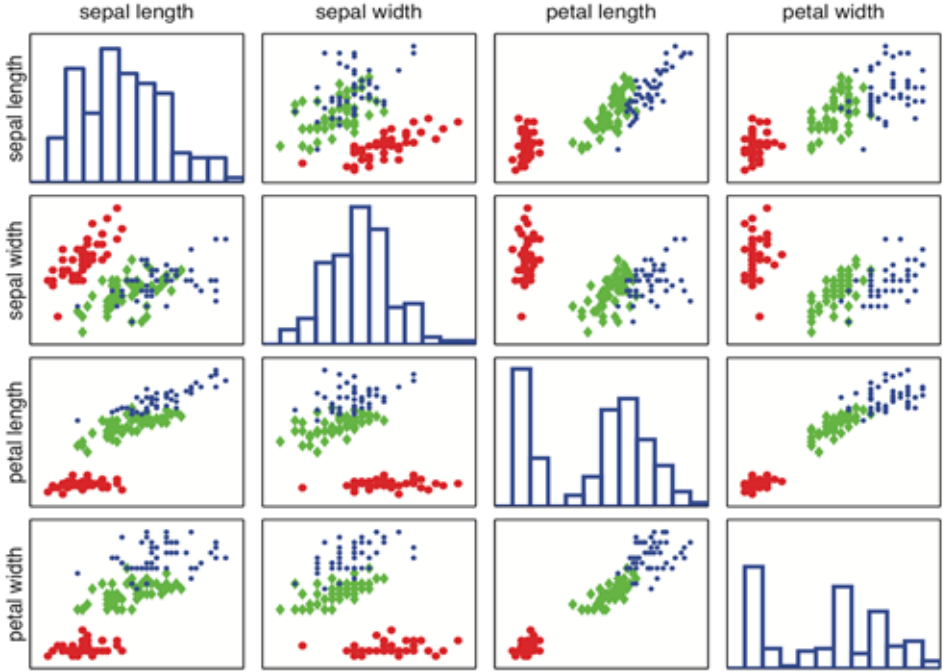


Figure 1.4 Visualization of the Iris data as a pairwise scatter plot. The diagonal plots the marginal histograms of the 4 features. The off diagonals contain scatterplots of all possible pairs of features. Red circle = setosa, green diamond = versicolor, blue star = virginica.

Classification: Emotion Detection

- Anger
- Disgust
- Fear
- Happiness
- Neutral
- Sadness
- Surprise

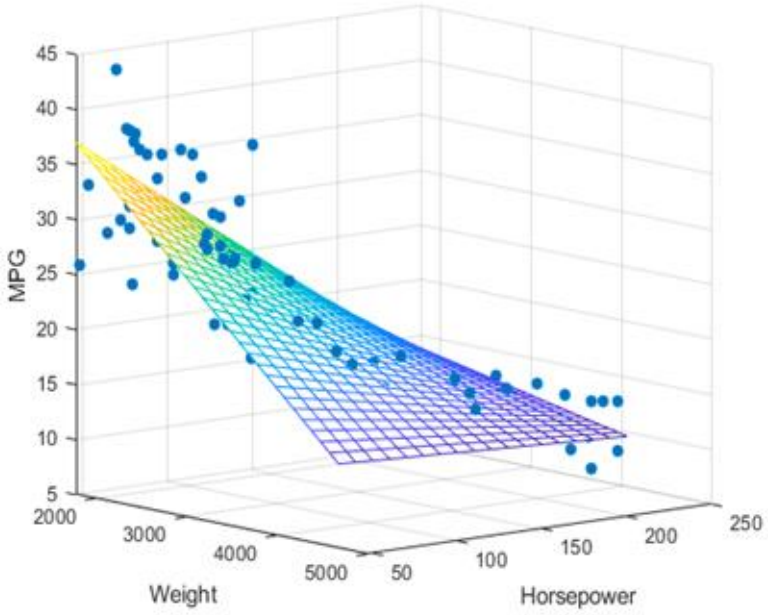




https://www.freecodecamp.org/news/facial-emotion-recognition-develop-a-c-n-n-and-break-into-kaggle-top-10-f618c024faa7/

Regression

- When the response variable is real-valued $y \in \mathbb{R}$



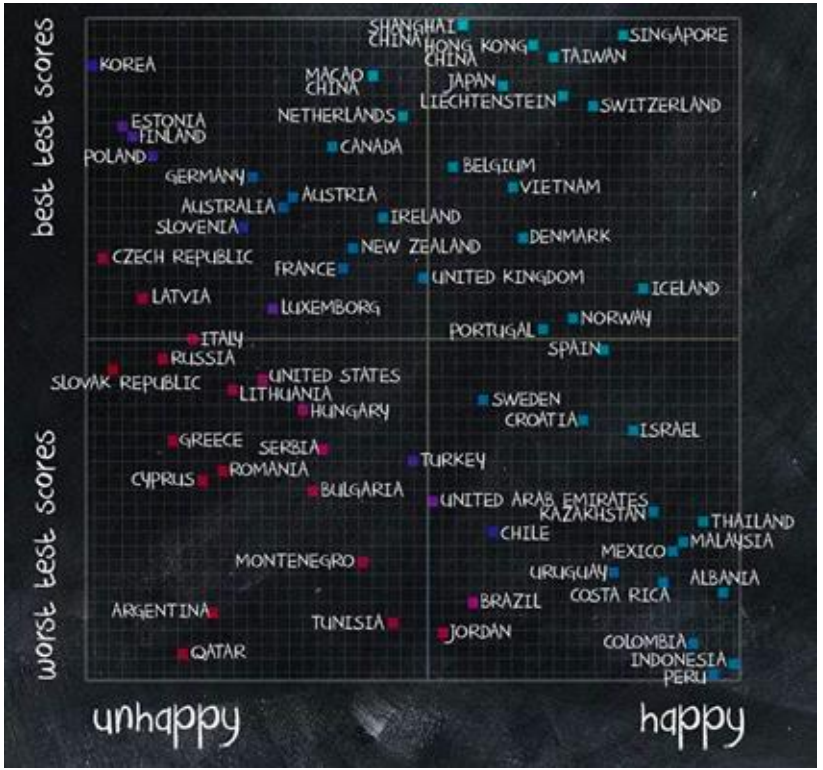


Unsupervised Learning

- Given a set of inputs $D = \{\boldsymbol{x}_i\}_{i=1}^{N}$
  - D is the input data
  - N is the number of examples
  - $\boldsymbol{x}$ is a d-dimensional vector of features/attributes/covariates
  - there is no response variable
- Learn a density function $p(\boldsymbol{x}_i|\theta)$
  - θ denotes the set of model parameters

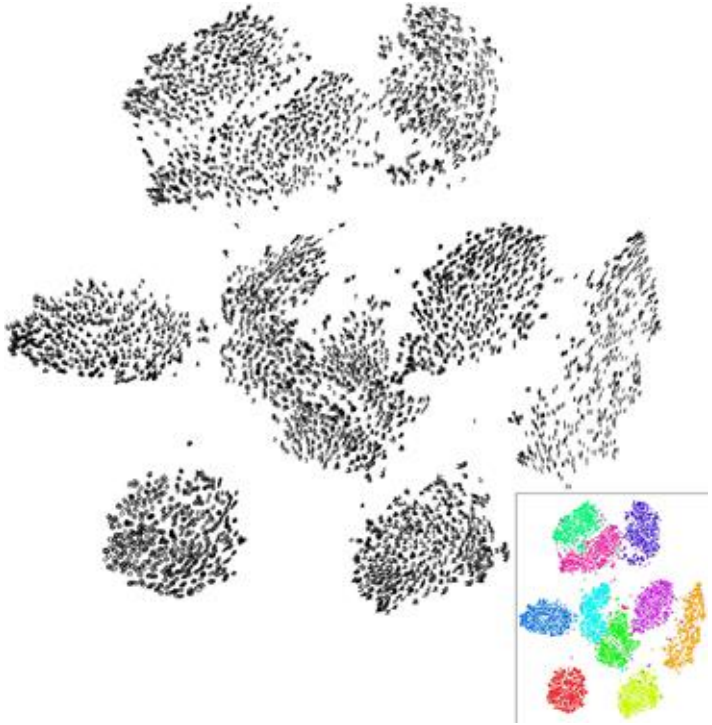
Clustering

- Clusters:
  - subgroups or subpopulations in the data
- Goals:
  - Discovering the subgroups
  - Estimating which subgroup a data point belongs to



https://i2.wp.com/techielobang.com/blog/wp-content/uploads/2014/01/best-school-happiest-kids.jpg

Clustering: MNIST Handwritten Digits



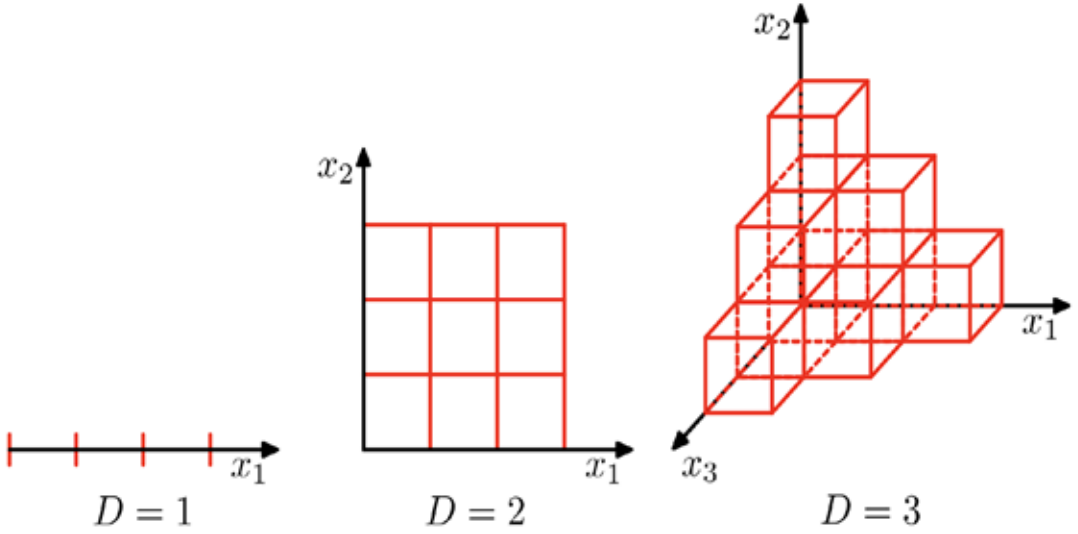
Source: http://lvdmaaten.github.io/tsne/examples/mnist_tsne.jpg

”The original black and white (bilevel) images from NIST were size normalized to fit in a 20x20 pixel box while preserving their aspect ratio. The resulting images contain grey levels as a result of the anti-aliasing technique used by the normalization algorithm. the images were centered in a 28x28 image by computing the center of mass of the pixels, and translating the image so as to position this point at the center of the 28x28 field.”

Source: http://yann.lecun.com/exdb/mnist/

Curse of Dimensionality

Figure 1.21 Illustration of the curse of dimensionality, showing how the number of regions of a regular grid grows exponentially with the dimensionality D of the space. For clarity, only a subset of the cubical regions are shown for $D = 3$.



Occam's Razor Principle

Occam's Razor is a principle that likes simplicity. It says that the simplest solution is usually the best one. In machine learning, this means that if we have two models that work about as well as each other, we should choose the simpler one.



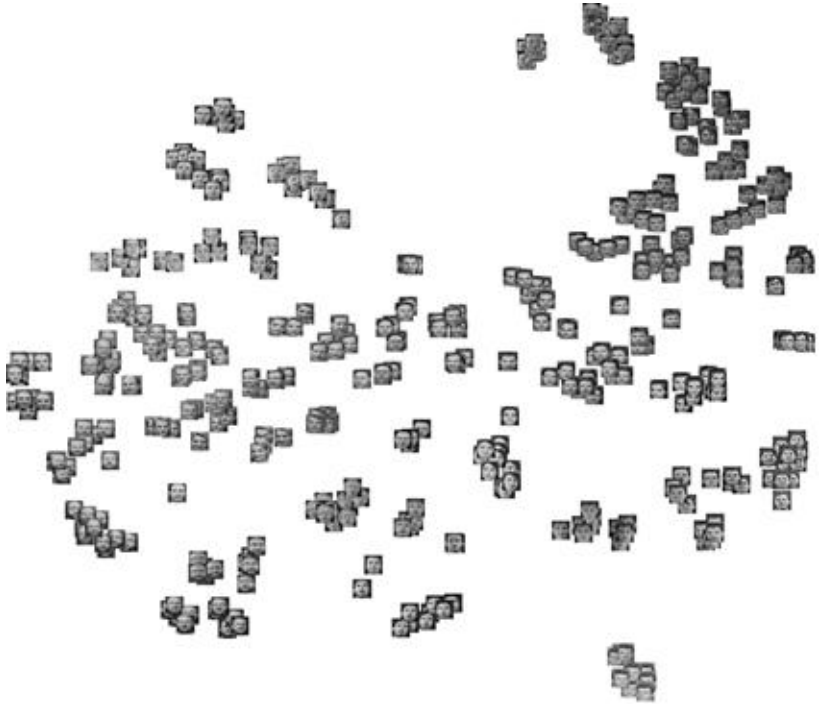
Dimensionality Reduction



Source:
https://www.flickr.com/photos/digitalanthill/4617998281/

Projecting the data to a lower-dimensional subspace that captures the “essence” of the data

Dimensionality Reduction: Olivetti Faces



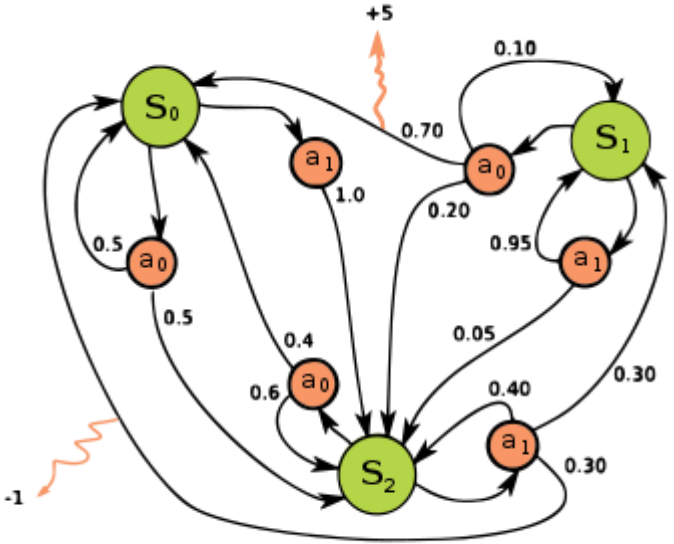
Source:
http://lvdmaaten.github.io/tsne/examples/olivetti_tsne.jpg

”There are ten different images of each of 40 distinct subjects. For some subjects, the images were taken at different times, varying the lighting, facial expressions (open / closed eyes, smiling / not smiling) and facial details (glasses / no glasses). All the images were taken against a dark homogeneous background with the subjects in an upright, frontal position (with tolerance for some side movement).”

Source:
https://www.cl.cam.ac.uk/research/dtg/attarchive/facedatabase.html

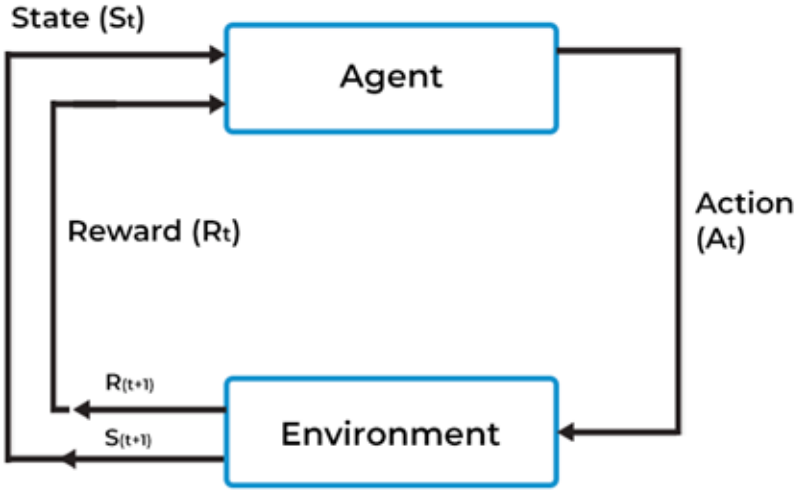
Markov Decision Process

Markov decision process (**MDP**) is a discrete-time stochastic control process. It provides a mathematical framework for modeling decision making in situations where outcomes are partly random and partly under the control of a decision maker



Source: https://en.wikipedia.org/wiki/Markov_decision_process

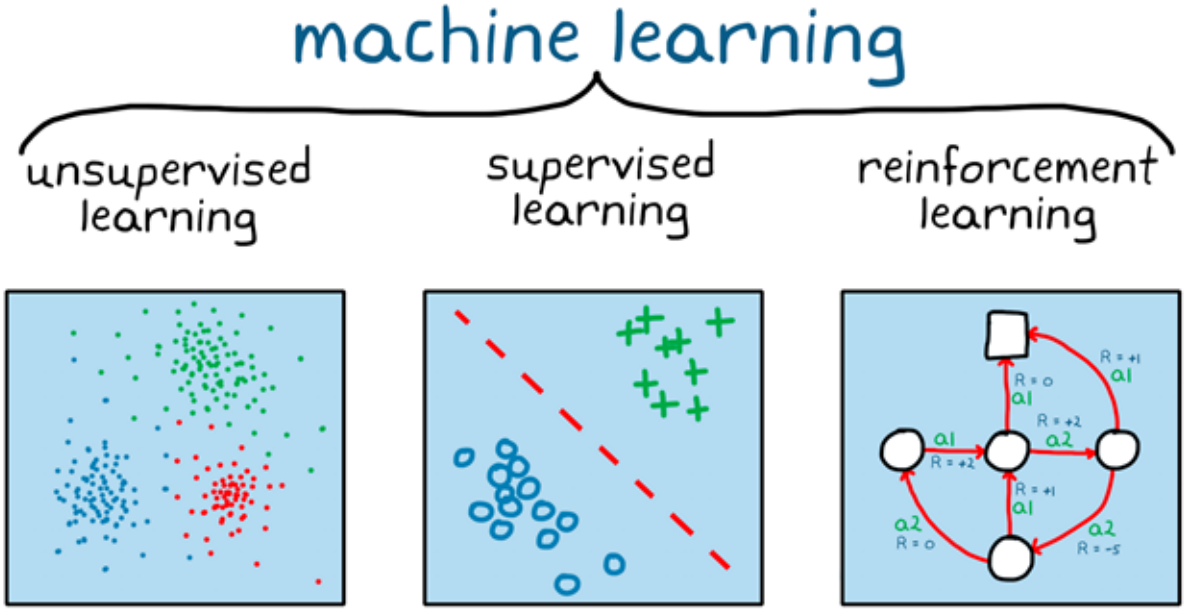
Reinforcement Learning



Reinforcement learning (**RL**) is an area of machine learning concerned with how intelligent agents ought to take actions in an environment in order to maximize the notion of cumulative reward

Source: https://en.wikipedia.org/wiki/Reinforcement_learning

Reinforcement Learning vs Traditional Learning



Source: https://www.mathworks.com/discovery/reinforcement-learning.html

Summary

- Introduction to Machine Learning
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  - Unsupervised Learning
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Q&A